A closure study to validate a novel isotope-aware Lagrangian cloud-microphysics model against airborne measurements

1 Scientific goal of the project

Water stable isotopologues are natural tracers of the global hydrological cycle. Variations in isotopic composition arise from isotopic fractionation accompanying phase transitions and transport processes in the atmosphere. Consequently, the isotopic composition of atmospheric water preserves information about the thermodynamic and microphysical history of clouds and precipitation.

In atmospheric sciences, the water isotopologues most commonly studied are HDO, $H_2{}^{18}O$, and $H_2{}^{17}O$. Their relative abundances differ from that of the dominant isotopologue $H_2{}^{16}O$, leading to measurable fractionation during evaporation, condensation, and mixing processes. Recent advances in cavity-enhanced laser spectroscopy have enabled continuous, high-precision measurements of multiple water isotopologues in atmospheric water vapour, including triple oxygen isotopes (^{16}O, ^{17}O, ^{18}O). The objective is to reconstruct cloud isotopic processes using numerical simulations. Measurements of the isotopic composition of water vapour below cloud base will be used as boundary and validation constraints for a particle-resolved cloud model. The simulated cloud state will subsequently be evaluated against observational data.

A central question is whether isotopic measurements provide sufficient information to constrain cloud microphysical processes and achieve *isotopic closure* within a prognostic particle-based framework. More broadly, it remains unknown whether isotopic closure can be achieved with current measurement capabilities and advances in particle-based cloud modelling. If so, an equally important question is what constitutes the minimal mathematical model, numerical implementation, and observational framework required to achieve it.

In this context, fluctuations in supersaturation play a key role in governing condensation and evaporation rates at the particle level, thereby influencing both cloud microphysics and isotopic partitioning. Recent work has shown that haze–cloud interactions can be strongly modulated by supersaturation variability (Fahandezh Sadi et al. 2026).

The project addresses the following research questions:

- **Q1** Can isotopic closure be achieved in a particle-based cloud model using currently available isotopic observations?
- **Q2** Which cloud microphysical processes leave identifiable signatures in the isotopic composition of atmospheric water?
- **Q3** To what extent do isotopic measurements constrain key cloud properties such as supersaturation variability, phase partitioning, and mixing?
- **Q4** What is the minimal physical and numerical framework required to achieve isotopic closure while maintaining predictive skill?

2 Significance of the project

Cloud microphysical processes remain one of the largest sources of uncertainty in weather prediction and climate projections (Intergovernmental Panel on Climate Change 2021). Processes such as condensation, evaporation, entrainment, and phase transitions between vapour, liquid, and ice strongly influence cloud lifetime, precipitation formation, and radiative properties. Yet many of these processes cannot be directly observed and therefore remain poorly constrained in atmospheric models (Grabowski and Wang 2013).

Stable water isotopologues provide a unique observational tracer of cloud evolution. During phase transitions, isotopic fractionation modifies the relative abundance of isotopologues in vapour, liquid, and ice (Lamb et al. 2017). Consequently, isotopic composition contains information about the thermodynamic and microphysical history of air masses and clouds (Bolot et al. 2013).

Theoretical descriptions of equilibrium and kinetic isotopic fractionation are well established, and stable water isotopologues are routinely represented in global and regional atmospheric models (Galewsky et al. 2016). These isotope-enabled models have significantly improved understanding of large-scale moisture transport and the atmospheric hydrological cycle (Risi et al. 2020). However, most existing approaches rely on Eulerian formulations and bulk cloud parameterisations, limiting their ability to resolve the particle-scale processes responsible for generating isotopic signatures (Joussaume et al. 1984; Yoshimura et al. 2008; Werner et al. 2011; Galewsky et al. 2016).

At the same time, particle-based cloud models have emerged as a state-of-the-art framework for studying cloud microphysics (Shima et al. 2009). By explicitly representing aerosol particles, cloud droplets, and ice particles, these models provide detailed descriptions of condensation, evaporation, freezing, and other microphysical processes.

Despite these advances, isotopic composition has rarely been considered within particle-based cloud modelling, and its potential as a constraint on cloud evolution remains largely unexplored. A fundamental unresolved question is whether isotopic observations contain sufficient information to constrain cloud microphysical processes and achieve isotopic closure. Analogously to cloud condensation nuclei closure studies, isotopic closure can be understood as the ability to consistently explain observed isotopic composition using a physically based description of cloud evolution (Knopf et al. 2021). While isotopic measurements are increasingly available and process-resolving cloud models are becoming increasingly sophisticated, no systematic framework currently exists for assessing isotopic closure in cloud systems.

The proposed project addresses this gap by investigating isotopic closure using particle-based cloud simulations and isotopic observations from in-cloud measurements and laboratory experiments. The project is pioneering because it introduces isotopic closure as a research objective in cloud physics and establishes a link between isotope-enabled observations and modern cloud microphysics. The results will provide insight into the information content of isotopic measurements, improve understanding of isotopic signatures generated by cloud processes, and contribute to the development of new observational constraints for atmospheric modelling.

3 Research Concept and Work Plan

The project is divided into three areas: development of a physically consistent theoretical description of isotopic fractionation in cloud microphysical processes (A1), its implementation in a particle-based cloud microphysics framework (A2), and evaluation against observational and laboratory data (A3). Feedback between these components ensures continuous refinement of both model formulation and interpretation of isotopic signals.

The proposed research builds on preliminary results obtained within a previous NCN project, in which the PI contributed to early development of isotope-enabled particle-based cloud modelling within the Super-Droplet Method framework. The preliminary results (PR) are described in each relevant research area (A1–A3) and provide a proof-of-concept for prognostic isotopic tracking in Lagrangian cloud microphysics.

A1 Theory: isotopic cloud microphysics

This research area focuses on developing a unified theoretical framework for isotopic fractionation in liquid–vapour systems, with a conceptual extension to ice-phase processes. The objective is to formulate governing equations that describe isotope-specific mass transfer in cloud environments, including both equilibrium and kinetic processes. The analysis will include: (i) derivation of isotopic mass balance equations for vapour, liquid, and ice phases, (ii) identification of limiting regimes such as equilibrium fractionation and Rayleigh-type distillation, and (iii) formulation of dimensionless control parameters, including the Bolin number introduced in preliminary studies.

A key objective is to identify the minimal set of physically essential processes that govern isotopic signatures in clouds. This includes condensation, evaporation, deposition, sublimation, and freezing, expressed in a form consistent with particle-based modelling. The outcome of A1 is a closed and consistent theoretical description of isotopic cloud microphysics suitable for numerical implementation.

In addition to model validation, the proposed framework includes a systematic analysis of numerical model errors, sensitivity to key physical parameters, and uncertainty propagation within the isotopic closure framework. This encompasses both local sensitivity analysis of governing equations and global sensitivity studies based on numerical experiments. Uncertainty propagation will be quantified through ensemble simulations, allowing assessment of how uncertainties in microphysical parameters, initial conditions, and isotopic boundary conditions influence the simulated isotopic composition. Particular attention will be given to distinguishing structural model uncertainty from observational uncertainty.

Preliminary results

A dimensionless framework for describing the evolution of isotopic composition of water in particle-based cloud models was introduced through the concept of the Bolin number. The formulation builds on the original work of Bolin (1958), developed for tritium transport in atmospheric water.

The Bolin number expresses the relative importance of diffusive and phase-change-driven isotopic exchange between isotopologues. It is defined as: $\mathbf{Bo} = \frac{\tau_{\text{heavy}}}{\tau_{\text{light}}}$, where $\tau = \left(\frac{1}{m}\frac{dm}{dt}\right)^{-1}$ is the characteristic equilibration timescale of a given isotopologue, and m denotes its mass in a droplet.

By defining this number explicitly, the evolution of a heavy isotopologue can be inferred from the dynamics of the corresponding light isotopologue (or approximately from the total droplet mass). The Bolin number provides a compact representation of isotopic exchange efficiency, incorporating equilibrium fractionation (α_{eq}), diffusive transport (including ventilation effects), and environmental controls such as temperature and saturation ratio. This formulation captures both equilibrium and kinetic fractionation regimes within a unified dimensionless framework.

The Bolin number was validated against equilibrium isotopic fractionation curves for water vapour and liquid phases reported by Gedzelman and Arnold (1994). The comparison is shown in Fig. 1, where colors denote values predicted by the Bolin-number-based formulation, while black lines reproduce the equilibrium regimes for liquid ($S^{\text{eq}}_{\text{liquid}}$) and vapour ($S^{\text{eq}}_{\text{vapour}}$) from Gedzelman and Arnold (1994).

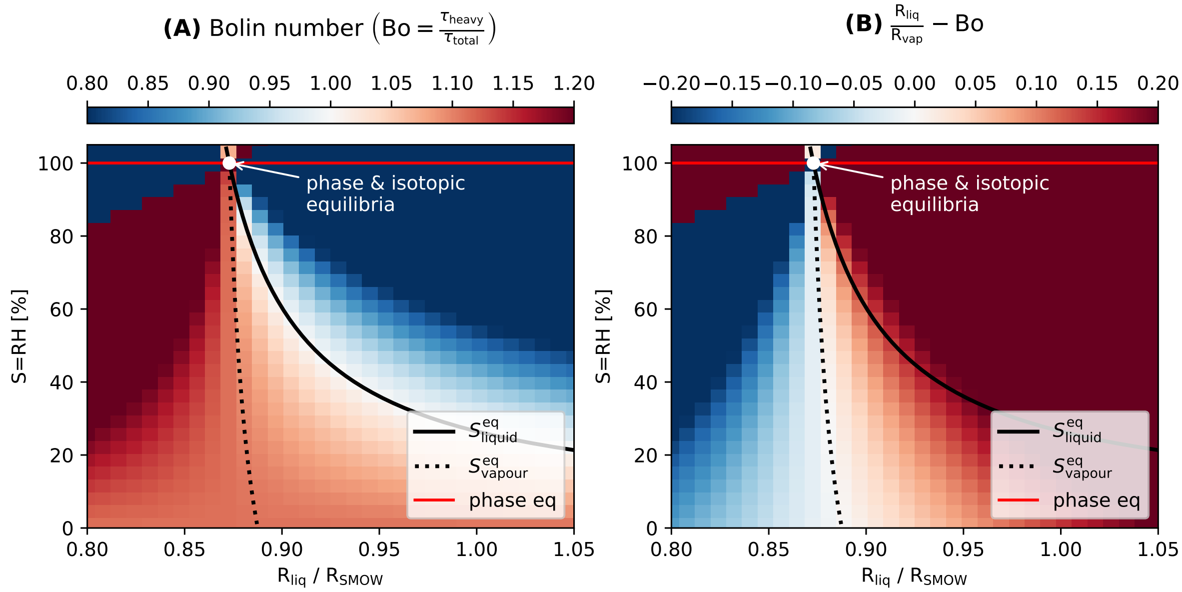

Figure 1: Conditions for isotopic steady state in liquid droplets (A) and ambient vapour (B). Panel A shows the locus where the isotopic composition of cloud droplets remains constant $((dR_{\mathrm{liq}}/dt = 0))$, corresponding to $(\mathrm{Bo} = 1)$. Panel B shows the locus where the isotopic composition of vapour remains constant $((dR_{\mathrm{vap}}/dt = 0))$, corresponding to $(R_{\mathrm{liq}}/R_{\mathrm{vap}} - \mathrm{Bo} = 0)$. The figure illustrates the theoretical equilibrium relationships governing isotopic exchange between liquid and vapour phases. The isotopologue considered is (HDO). Black lines indicate isotopic equilibrium curves, the red line phase equilibrium (RH = 100%), and the white dot simultaneous phase and isotopic equilibrium.

The developed concept has been accepted for presentation at the 17th Conference on Cloud Physics (AMS Madison Summit 2026).

A2 Numerics: particle-based implementation and simulation framework

This research area focuses on implementing the proposed theoretical framework within the particle-based cloud microphysics model PySDM, which is based on the Super-Droplet Method (SDM) (Shima et al. 2009).

State of the art

The Super-Droplet Method (SDM) is a particle-based Lagrangian cloud microphysics framework in which cloud and aerosol populations are represented by computational particles (super-droplets) carrying multiplicities (Shima et al. 2009).

PySDM is an open-source implementation of SDM that enables simulations of cloud microphysical processes such as condensation, evaporation, collision–coalescence, and ice-phase transformations in box, parcel, and single-column configurations (Bartman et al. 2022; de Jong et al. 2023). Its particle-based structure allows straightforward extension with additional prognostic variables, such as isotopic composition.

Preliminary results

The PI has implemented isotopic evolution of water vapour and cloud droplets for a single heavy isotopologue within the PySDM framework based on the Super-Droplet Method. This extension of PySDM enables prognostic isotopic composition of individual computational particles, allowing explicit representation of isotopic fractionation during cloud microphysical processes.

Planned work

Building on the existing conceptual formulation, the model will be implemented within PySDM by extending the current single-isotopologue treatment to a fully mixed-phase, multi-isotopologue framework. At present, isotopic tracking has only been developed for vapour and liquid phases; representation of isotopes in the ice phase remains to be implemented. The proposed work will therefore generalise the formulation to include prognostic isotopic composition in all three phases and enable consistent treatment of fractionation during phase transitions, diffusive exchange, and collisional processes within a particle-resolved framework.

The numerical implementation will be verified using conservation diagnostics and comparison against analytical limiting cases. After implementation, the model will be explored in idealised numerical experiments (box, parcel, and column configurations) to quantify sensitivity of isotopic composition to key microphysical and environmental controls, including supersaturation, temperature, droplet size distribution, and ice fraction.

The outcome will be a fully mixed-phase, isotope-enabled PySDM extension suitable for subsequent application to cloud process studies.

A3 Data: observational and experimental validation

This research area is dedicated to assessing model predictions through comparison with observational and experimental data. The model will be tested using isotopic measurements from aircraft campaigns, including the CAESAR dataset, as well as laboratory experiments performed in collaboration with NSF NCAR (Elise Rosky). Together, these datasets offer complementary constraints spanning vapour, cloud water, ice, and precipitation phases.

The laboratory component is currently in a preliminary stage, and its design will evolve in parallel with the development of the modelling framework. This co-development approach highlights the potential and scientific importance of the proposed project, as it enables direct feedback between numerical modelling and experimental design. It also ensures consistency between observational capabilities, experimental constraints, and the evolving requirements of the analysis, while supporting close interdisciplinary collaboration.

The analysis will focus on three aspects: (i) agreement between simulated and observed isotopic composition, (ii) sensitivity of isotopic signals to cloud microphysical processes, and (iii) ability of isotopic measurements to constrain cloud evolution and assess closure. The outcome of A3 is an empirical evaluation of isotopic closure and a quantification of the information content of isotopic observations in mixed-phase cloud systems.

Data availability

The CAESAR field campaign provides an openly accessible dataset of cold-air outbreak clouds over the Nordic Seas (EOL archive). Collaboration with Elise Rosky was initiated during the PI's visit to NSF NCAR in 2025. As an outcome of this collaboration, the PI contributed to a conference abstract presented by Dr. Rosky at the AGU Fall Meeting 2025 on isotopic observations from the CAESAR campaign (Rosky et al. 2025). The dataset will serve as one of the observational benchmarks for model evaluation within the proposed project.

Integration of A1–A3

The three research areas are strongly coupled. A1 defines the physical description of isotopic processes, A2 translates this description into a numerical particle-based framework, and A3 provides observational constraints and evaluates model performance against measurements and laboratory data. This integrated structure enables a systematic investigation of isotopic closure, ensuring consistency between theory, numerical representation, and observational evidence.

4 Project Tasks, Milestones and Schedule

The task structure follows the three research areas defined in the project concept (A1, A2, A3), ensuring consistency between theoretical development, numerical implementation, and data-driven evaluation. This structure supports iterative feedback between model formulation and physical interpretation of isotopic signals.

The research plan is formulated according to the SMART principle (specific, measurable, achievable, relevant, and time-bound). Each work package is decomposed into specific tasks.

Tasks

The project consists of core tasks structured into three work packages: theoretical development, numerical implementation, and observational validation. Each task produces at least one major research output (peer-reviewed publication and/or open-source software release).

A1 Theory: Isotopic Cloud Microphysics

T1 Derive a closed set of governing equations for isotope-resolved mass transfer between vapour, liquid, and ice phases in mixed-phase clouds. The formulation includes equilibrium fractionation, kinetic effects, and phase-change-driven non-equilibrium processes in a particle-based framework.

Deliverable: Results presented on EGU General Assembly (Months 6–12).

T2 Identify limiting regimes of isotope evolution (equilibrium, Rayleigh distillation, diffusion-limited growth) and construct reduced-order models. Develop a regime map of isotopic behaviour and sensitivity structure.

Deliverables: (i) reduced-order model suite with validity regimes clearly defined, (ii) diagnostic criteria for isotopic closure breakdown.

Dissemination: Conference poster/talk (EGU) (Months 12–18).

T3 Quantify sensitivity of isotopic composition to microphysical parameters and atmospheric forcing using local and global sensitivity analysis, and construct an uncertainty propagation framework linking parameter uncertainty to isotopic observables.

Deliverables: (i) ensemble-based uncertainty quantification framework, (ii) reproducible analysis notebooks.

Dissemination: ICCP / EGU presentation (Months 18–24).

A2 Numerical Framework: Particle-Based Implementation

T4 Implement isotope-resolved cloud microphysics in the PySDM framework, including vapour, liquid, and ice phases. Ensure mass conservation, thermodynamic consistency, and validation against analytical limiting cases.

Deliverable: Open-source PySDM isotope module (Zenodo DOI + GitHub release).

Dissemination: EGU demo (Months 12–18).

T5 Develop a minimal isotope-enabled cloud model for controlled numerical experiments. The framework isolates key processes controlling isotopic closure, including supersaturation variability, mixing, and phase partitioning.

Deliverables: (i) full mixed-phase isotope-enabled PySDM extension, (ii) verification against analytical fractionation limits, (iii) documented API for isotopic particle attributes.

Dissemination: Summer school/tutorial release (Months 14–20).

A3 Validation and Closure Assessment

T6 Construct and preprocess observational and laboratory datasets (CAESAR, NCAR experiments), including uncertainty characterization and mapping to model state variables.

Deliverable: Curated, model-ready dataset + preprocessing pipeline.

T7 Perform model–observation comparison and evaluate isotopic closure. Identify which microphysical processes are required to reproduce observed isotopic signatures and quantify closure regimes.

Deliverables: (i) formal closure assessment framework, (ii) classification of closure regimes (closed / partially constrained / unconstrained).

Dissemination: Peer-reviewed paper on isotope closure, final open-source release of full framework (model + diagnostics + datasets) (Months 18–24).

Milestones

M1 Completion of theoretical framework for isotope-resolved cloud microphysics and identification of governing equations (Month 12).

M2 Completion of numerical implementation in PySDM and validation of isotope tracking in idealised simulations (Month 18).

M3 Integration of observational and laboratory datasets with model framework and initiation of systematic evaluation (Month 12).

M4 Completion of isotopic closure assessment and identification of minimal physical requirements for closure (Month 24).

Schedule

The project is structured over 24 months with strong feedback between theory, numerics, and data analysis. Dataset preparation starts early, while closure analysis is performed in the final phase.

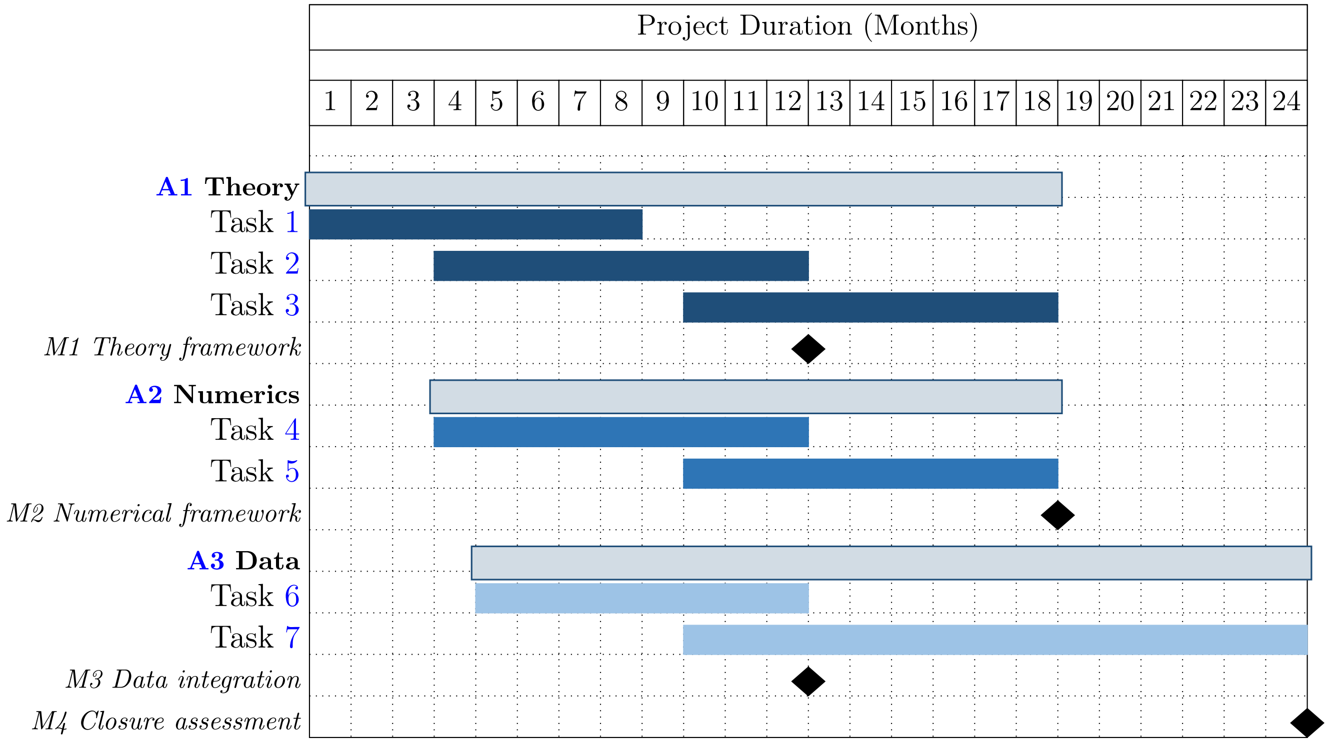

Figure 2: Project timeline structured into three work packages: A1, A2, and A3.

Result dissemination

Project results will be disseminated through peer-reviewed publication, conference presentations, research visits, and open-source software releases. Results will be submitted to international journal relevant to atmospheric modelling and isotope hydrology. Journals taken into account: *Tellus B: Chemical and Physical Meteorology*, *Journal of Advances in Modeling Earth Systems*, and *Geoscientific Model Development*. Interim and final results will be presented as talks or posters at international conferences, including the European Geosciences Union General Assembly and the International Conference on Clouds and Precipitation.

The project includes a research visit to the United States, a visit of an international collaborator to AGH University of Krakow, and participation of the MSc student in an international summer school. All software developed within the project will be released as open source via GitHub, accompanied by documentation and reproducible Jupyter notebooks demonstrating the proposed methods and numerical experiments. Data-processing workflows required to reproduce published results will be made publicly available.

5 Research methodology

Underlying scientific methodology, methods, techniques and research tools, methods of results analysis, and equipment used in the research are based on a combination of physically based modelling, particle-based numerical simulation, and systematic data-model comparison.

PySDM-based modelling framework

The numerical core of the model is the PySDM implementation of the Super-Droplet Method (SDM), which is used as the Lagrangian particle-resolved framework for cloud microphysics in this project. In this work, PySDM is extended by augmenting each super-droplet with prognostic variables describing the isotopic composition of water in vapour, liquid, and ice phases. This preserves the original SDM structure while enabling simulation of isotope-resolved cloud microphysics. The resulting framework is used for idealised box, parcel, and column simulations to study isotope–microphysics interactions under controlled conditions.

Jupyter notebook–based simulation and analysis environment

A central component of the project is the use of Jupyter notebooks as the primary interface for model development, numerical experiments, and data analysis. Each notebook constitutes a fully self-contained and executable research workflow, including (i) model configuration and initialization, (ii) simulation execution, (iii) post-processing and diagnostics, and (iv) visualisation of results. This design ensures that every numerical experiment is fully reproducible and can be independently executed without additional external scripts or hidden dependencies.

Notebooks function simultaneously as an executable research workflow, a documentation format, and a reproducible dissemination layer for scientific results. Within this project, notebooks are integrated with version control and continuous integration (CI) pipelines. This ensures that all simulations remain reproducible over time and that changes in the codebase automatically trigger validation of numerical correctness.

Reproducibility, software engineering, and research workflow

The project follows established reproducible research practices. All components of the modelling system, including PySDM extensions, simulation configurations, and analysis notebooks, are managed in version-controlled repositories and continuously validated through automated CI pipelines. The CI system performs unit tests verifying physical consistency (e.g. mass conservation and isotopic bounds), regression tests ensuring numerical stability, and benchmark tests against analytical or well-established reference solutions. This workflow ensures that model development remains transparent, scientifically traceable, and computationally robust.

Educational and training component

The Jupyter-based workflow plays a key role in the training component of the project. The MSc/Erasmus+/PhD student involved in the project will contribute directly to the development of simulation notebooks, including simplified educational model configurations, reproducible experiment templates for key processes, and interactive visualisation and analysis tools.

This approach lowers the entry barrier to particle-based cloud microphysics and provides hands-on training in modern scientific computing, including numerical modelling, uncertainty quantification, and reproducible research practices. As a result, the project integrates scientific development with structured training in computational atmospheric physics.

Composition and qualifications of the research team

The research team consists of the Principal Investigator (PI), a scientific mentor, and one student, forming a compact group focused on particle-based cloud microphysics and isotope-enabled modelling.

Principal Investigator The Principal Investigator is a PhD student in the physical sciences with a background in mathematics and numerical modelling. The PI has experience in scientific computing and Python-based development, with a focus on particle-based atmospheric simulation frameworks. The PI is the main developer of the isotope-enabled cloud microphysics framework presented in this proposal, integrating theoretical formulations, numerical implementation, and data analysis within a unified modelling environment. Previous work includes isotope-enabled cloud modelling and international collaboration at the National Center for Atmospheric Research (NCAR), including work with Dr. Elise Rosky on mixed-phase cloud isotopic processes.

Scientific mentor The scientific mentor is Dr. Sylwester Arabas, co-developer of the PySDM particle-based cloud microphysics framework and an expert in Lagrangian cloud modelling and scientific computing. The mentor contributes to the development of open-source atmospheric modelling tools and ensures the physical and numerical robustness of the SDM-based framework.

Student contribution The student (MSc/Erasmus+/PhD level) will support development of a minimal computational environment for isotope-enabled cloud simulations. The implementation will be in Python (optionally Julia) and will serve as a reusable and accessible framework for research and training in cloud microphysics.

Environmental Physics group background The PI and scientific mentor are affiliated with the Environmental Physics group, which conducts research on atmospheric isotope systematics and tracer-based hydrological processes. The group has experience in the measurement, analysis, and interpretation of stable water isotopes in environmental and atmospheric applications, including diffusion and phase-change fractionation processes (Pierchała et al. 2022). This background supports the integration of isotope observations with particle-based cloud modelling and provides methodological continuity between observational and numerical approaches.

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